COMPOSITE OF STANDARD V COVERINGS OF AFFINES IS A STANDARD V COVERING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [3718, 3724] and proof(s) [[3726, 3731]]; this spec covers the complete environment instead (statement 3718-3724, proof span(s) [(3726, 3731)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Lemma 0.1. *Let T be an affine scheme. Let $\{T_j \to T\}_{j=1,\ldots,m}$ be a standard V covering. Let $\{T_{ji} \to T_j\}_{i=1,\ldots n_j}$ be a standard V covering. Then $\{T_{ji} \to T\}_{i,j}$ is a standard V covering.*

Proof. This follows formally from the observation that if $V \subset W$ and $W \subset \Omega$ are extensions of valuation rings, then $V \subset \Omega$ is an extension of valuation rings. $\square$

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